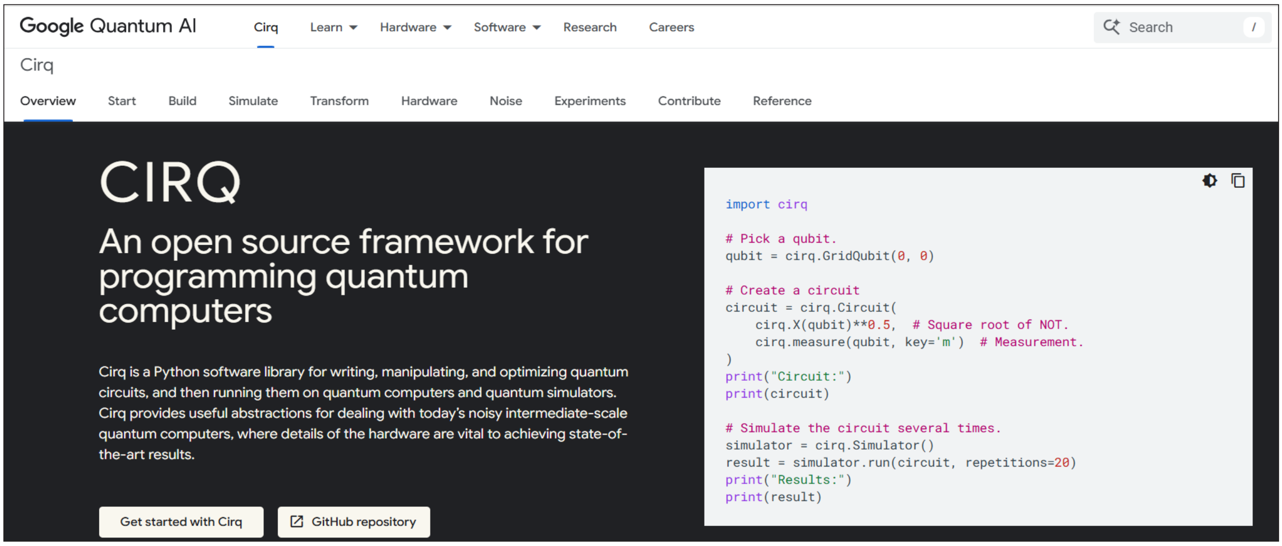

Figure 3: Cirq framework for quantum information processing and simulation of QML

To install Pennylane, use the following code:

```
$ pip install pennylane

from pennylane import numpy as np
import pennylane as qml
import pandas as pd

# Sample dataset
mydata = pd.DataFrame({
    "Age": [27, 36, 35],
    "PurchasedItem": [0, 1, 1]
})
print(mydata)
# Integration of Quantum simulator
mydev = qml.device("default.qubit", wires=1)

@qml.qnode(mydev)
def quantum_featuremap(x):

    # Input features encoding
    qml.RX(x/50.0, wires=0)

    return qml.expval(qml.PauliZ(0))

# Quantum Transformation on Sample Data
mydata["Quantum_Feature"] = mydata["Age"].apply(quantum_
featuremap)

print(mydata)

OUTPUT
   Age  PurchasedItem
0   27              0
1   36              1
2   35              1

   Age  PurchasedItem  Quantum_Feature
0   27              0         0.857709
1   36              1         0.751806
2   35              1         0.764842
```

The Quantum_Feature you get after executing and simulating the quantum circuit presents the quantum encoding. The quantum circuits work as mathematical transformers for data analytics and data engineering

Google Cirq

URL: https://quantumai.google/cirq

Google Cirq is an open source framework that can be used for the simulation of quantum circuits for quantum data engineering and quantum machine learning.

It has a range of features for experimenting and research in quantum machine learning and quantum data science. Cirq helps developers design, implement, simulate and optimize quantum circuits using Python programming.

Quantum computing-based cloud simulators can be used to solve a range of real world problems. IBM Qiskit, Google Cirq, PennyLane and many other frameworks provide quantum circuits that can be simulated on the cloud, Google Colab notebooks or on local devices to get effective and accurate solutions in areas being researched. END

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The author is associated with various academic and research institutes for delivering expert lectures and conducting technical workshops on the latest technologies and tools.